

Oct-22/TE/Insem-641
T.E. (Artificial Intelligence and Machine Learning)
PATTERN RECOGNITION
(Elective-I) (2019 Pattern) (Semester - I) (318545B)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is Pattern Recognition? Explain the application of Pattern Recognition. [5]
- b) What is features and feature vectors? [5]
- c) Explain Gaussian Probability density function in details. [5]

OR

- Q2)** a) Differentiate between Supervised and unsupervised learning with the help of example. [5]
- b) Explain Bayesian classification for normal distribution. [5]
- c) Explain the term decision region and decision boundaries. [5]
- Q3)** a) Write a short note on SVD. [5]
- b) What is vector? Explain the types of vector. [5]
- c) Write a short note on KL transform. [5]

OR

- Q4)** a) State the difference between linear and nonlinear Dimensionality Reduction. [5]
- b) What is principle component analysis? Explain with the help of example. [5]
- c) What is data transformation? Explain the types of data transformation. [5]

